

LECTURE 9

Computational Fluid Dynamics: Foundations

Reference: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Chapters 1 and 2.

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Computational Fluid Mechanics/Dynamics

The processes of fluid mechanics and heat transfer are governed by partial differential equations (PDEs).

CFD solves those equations numerically.

The continuum is replaced by a finite set of **points or volumes**

Each derivative is replaced by an algebraic expression in the values at those points.

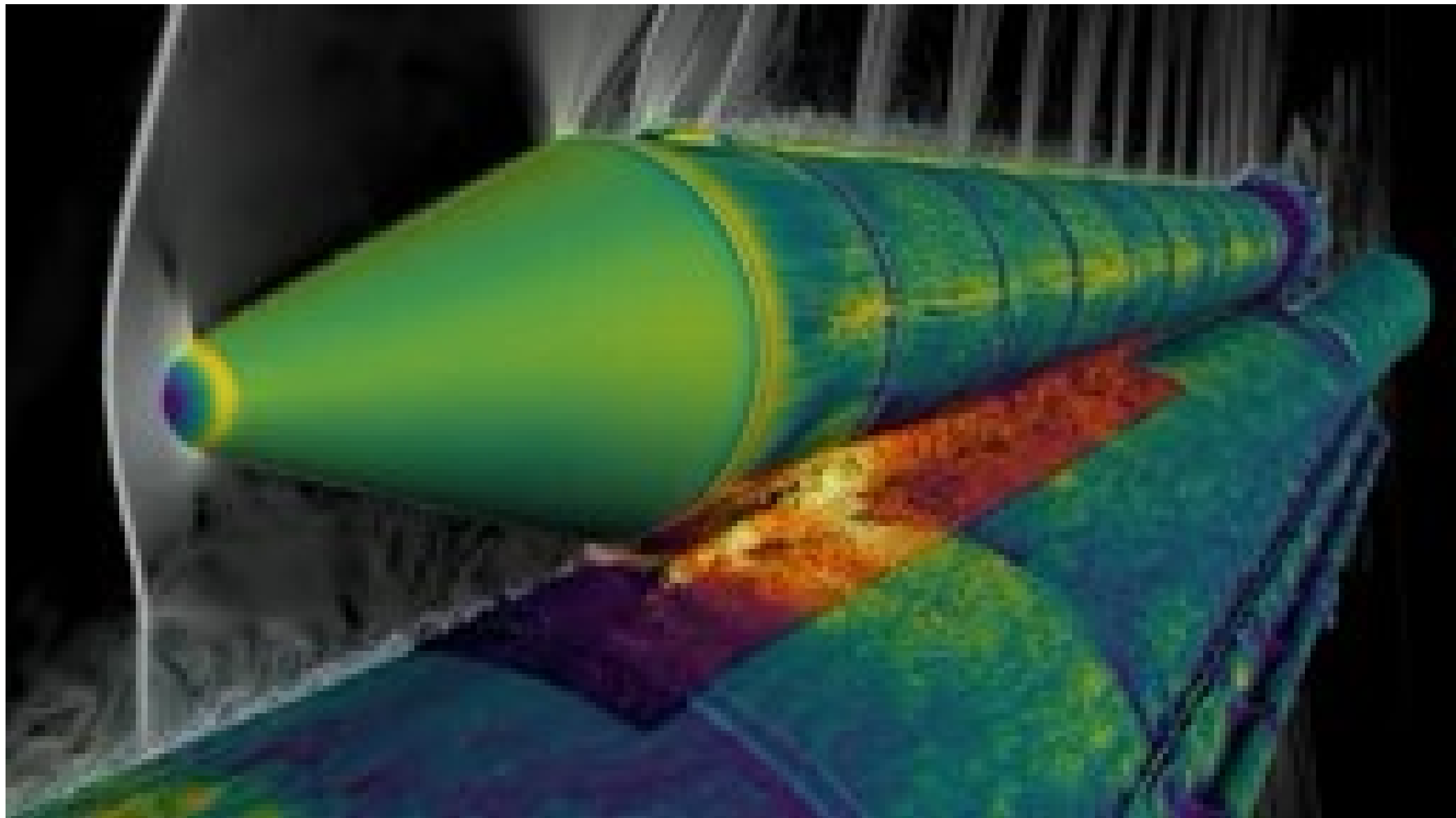
The output is generally tabulated data, where we have one value of each variable at each point of the grid, at each instant.

For CFD, 3 ingredients are required:

- The governing equations
- The numerical methods
- A computer.

What a modern CFD calculation looks like

A present-day simulation resolves the flow around a complete object with tens of millions of cells, on thousands of processors.



NASA Artemis II Rocket Supercomputer Simulation

<https://www.youtube.com/watch?v=oJqk26YyKRg>

Three approaches to a flow problem

Experimental:

A scale model in a facility able to reproduce the required free-stream conditions, instrumented and measured.

Theoretical:

Simplifying assumptions carried as far as a closed-form solution, valid for simple geometry and usually linear.

Computational:

Few assumptions, and the governing equations solved numerically on a grid.

The three are complementary, not competing.

Turbulent, multiphase and reacting flows still depend on experiment.

Computation removes the constraint that the test conditions must be reproducible in a laboratory -> Numerical experiments

Advantages and disadvantages

Table 1.1 Comparison of approaches

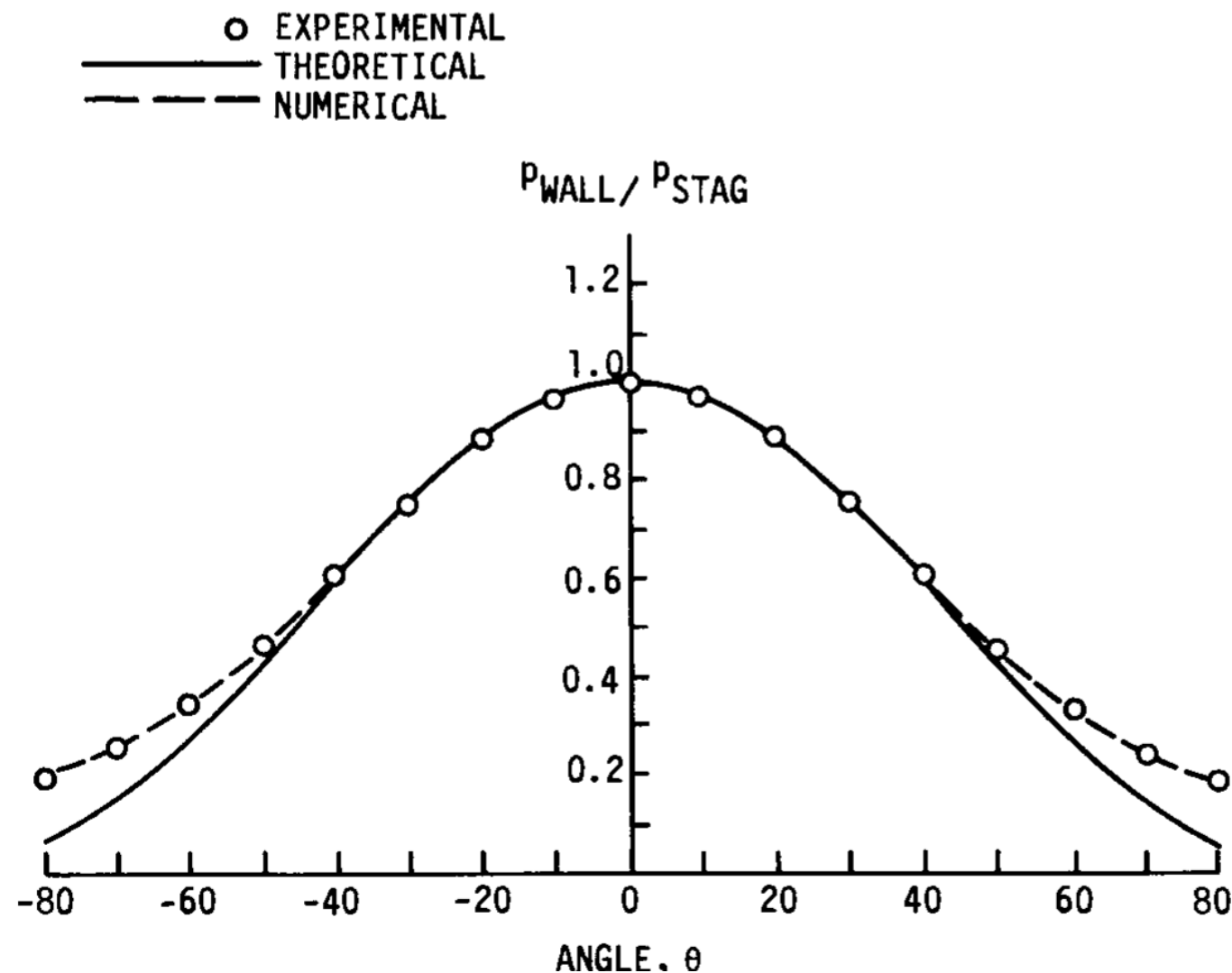
Approach	Advantages	Disadvantages
Experimental	1. Capable of being most realistic	1. Equipment required 2. Scaling problems 3. Tunnel corrections 4. Measurement difficulties 5. Operating costs
Theoretical	1. Clean, general information, which is usually in formula form	1. Restricted to simple geometry and physics 2. Usually restricted to linear problems
Computational	1. No restriction to linearity 2. Complicated physics can be treated 3. Time evolution of flow can be obtained	1. Truncation errors 2. Boundary condition problems 3. Computer costs

Comparison of the three approaches. Tannehill, Anderson & Pletcher.

The computational row does not claim to be accurate.
It claims to be unrestricted.

Comparison of the 3 approaches

Supersonic flow over a circular cylinder at Mach 4 and Reynolds number 5×10^5 , treated by each of the 3 methods.



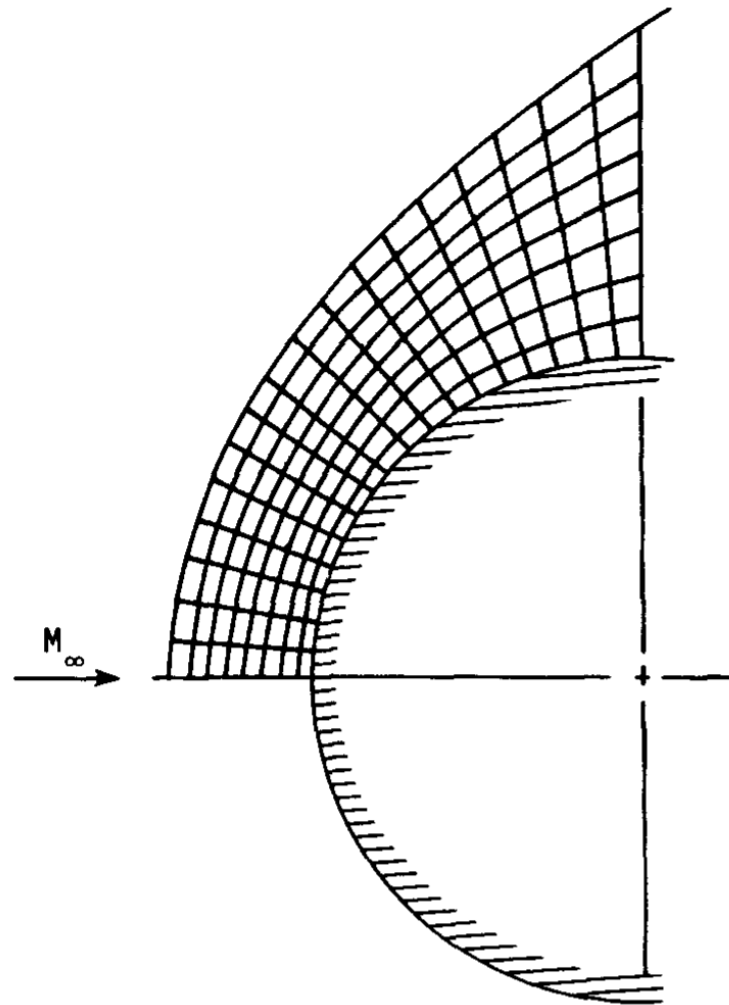
Surface pressure on a circular cylinder. Tannehill, Anderson & Pletcher

Modified Newtonian theory holds to about $\pm 35^\circ$.

The numerical solution of the Euler equations follows the measurements all the way round.

The computational grid

The region between shock and body is subdivided.
Each partial derivative becomes a finite difference at each grid point.

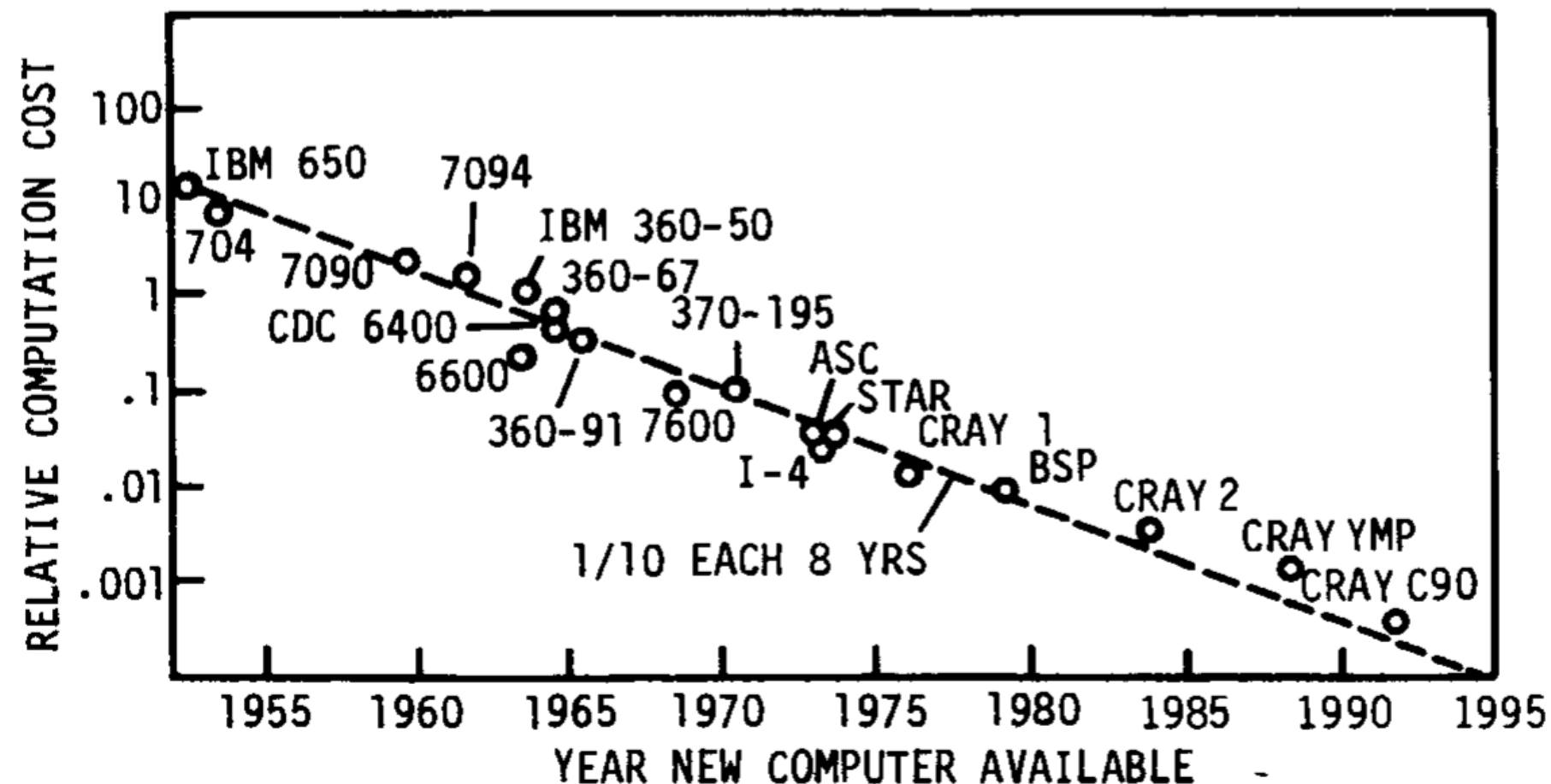


Computational grid for the blunt-body problem. Tannehill, Anderson & Pletcher

The resulting algebraic equations are integrated forward in time until the solution stops changing: the steady state is reached asymptotically.

The economics of computing

The cost of a given calculation has fallen by about a factor of ten every eight years. The cost of experiment has risen.



Relative computation cost for a given flow and algorithm. Tannehill, Anderson & Pletcher, Fig. 1.1, p. 4.

Part of the gain is hardware and part is better algorithms.

A better algorithm keeps its advantage on every machine that follows.

The economics of computing

Same trend is seen today:



Historical milestones: before computers

1910, Richardson:

point iterative solution of Laplace's equation, by hand, for the stress distribution in a masonry dam.

1918, Liebmann:

the same sweep, but each value is updated as soon as it is computed. Convergence is considerably faster.

1928, Courant, Friedrichs and Lewy:

Existence and uniqueness for difference solutions; the origin of the CFL condition.

1940, Southwell:

Relaxation driven by the largest residual, designed for hand calculation.

Richardson also drew the distinction we are about to make:

problems that must be relaxed, and problems that can be marched.

Historical perspective: the first machines

1950, O'Brien, Hyman and Kaplan:

The first full account of von Neumann's stability analysis, which he never published.

1950, Frankel:

Successive overrelaxation: a large gain in convergence rate for elliptic problems.

1954, Lax:

Shock capturing. The conservation-law form allows discontinuities with no special treatment.

1955, Peaceman and Rachford:

Alternating direction implicit methods, with no restriction on the step size.

Von Neumann's stability analysis remains the standard tool.

Historical perspective: shocks and upwinding

1959, Godunov:

A Riemann problem solved at every cell face to supply the flux.

1960, Lax and Wendroff:

Second-order accuracy without the smearing of the earlier schemes; MacCormack's version, 1969.

1971, Murman and Cole:

Type-dependent differencing: the scheme changes where the equation changes type.

1979-80, Steger and Warming; Roe.

Flux splitting, and an approximate Riemann solver cheap enough to use.

These ideas are the core of every production code written since, and the reason upwinding is the default for convection.

Historical perspective: limiters and multigrid

1973, Boris and Book:

Flux correction: the first nonlinear limiter on the numerical flux.

1983, Harten:

Total variation diminishing schemes, which generalise limiting and forbid new extrema.

1962-1977, Fedorenko and Brandt:

The multigrid method, first for elliptic equations and then for fluid flow.

1980s onwards:

The finite-volume formulation and unstructured grids make complete configurations tractable.

Nonlinear limiting with controlled dissipation is the defining feature of a modern shock-capturing scheme.

Partial differential equations (PDEs)

Before choosing a numerical method we must ask a prior question:
what kind of equation is being solved?

Two classifications are used:

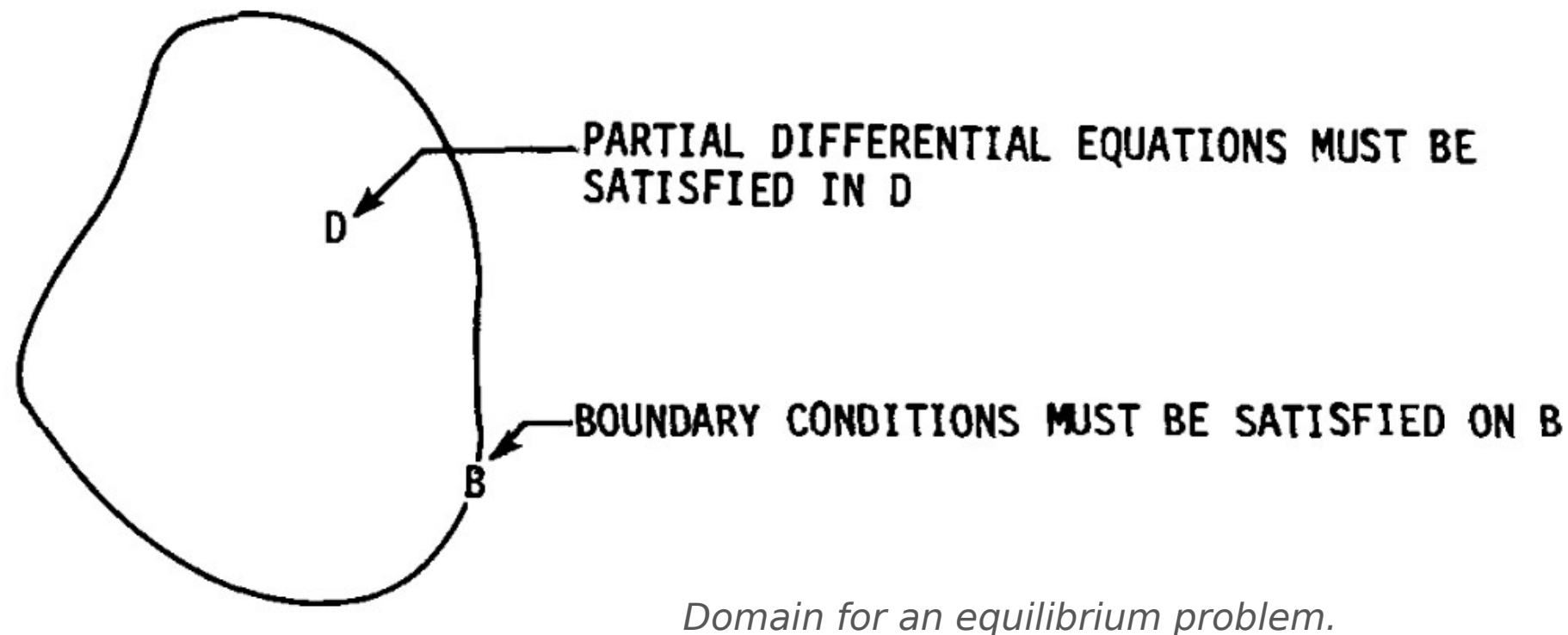
1. The physical one separates **equilibrium problems** from **marching problems**.
2. The mathematical one rests on characteristics, and gives the **elliptic**, **parabolic** and **hyperbolic** types.

The 2 classifications agree:

- Equilibrium problems are elliptic
- Marching problems are hyperbolic or parabolic.

Equilibrium problems

The solution is required in a closed domain D , subject to conditions prescribed on the whole of the boundary B .



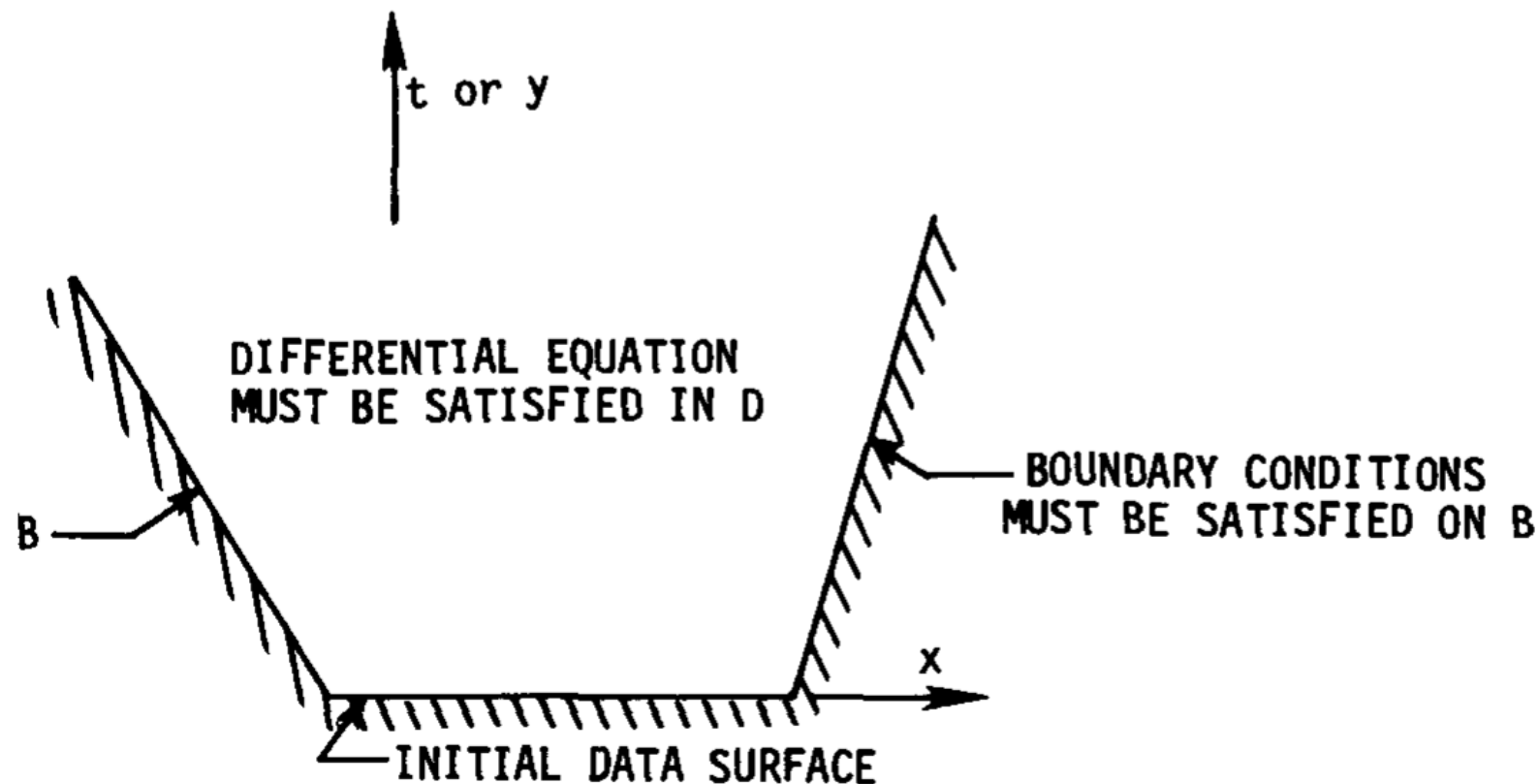
These are **boundary value problems**, also called jury problems: the solution at every interior point depends on every point of the boundary.

Examples:

Steady temperature distributions, incompressible inviscid flow, equilibrium stress in a solid.

Marching problems

The solution is required on an open domain, computed outward from an initial data surface while the boundary conditions are satisfied.



Domain for a marching problem.

These are **initial value** or **initial-boundary value problems**.

The marching direction is time, or a coordinate that behaves like time.

Examples:

Unsteady inviscid flow, steady supersonic flow, transient conduction, boundary-layer flow.

The general second-order equation

Take a linear second-order equation in two independent variables, with coefficients that depend on x and y .

$$a \frac{\partial^2 \phi}{\partial x^2} + b \frac{\partial^2 \phi}{\partial x \partial y} + c \frac{\partial^2 \phi}{\partial y^2} + d \frac{\partial \phi}{\partial x} + e \frac{\partial \phi}{\partial y} + f \phi = g(x, y)$$

The classification depends only on the second-derivative terms, so everything else is collected on the right-hand side.

$$a \frac{\partial^2 \phi}{\partial x^2} + b \frac{\partial^2 \phi}{\partial x \partial y} + c \frac{\partial^2 \phi}{\partial y^2} = - \left(d \frac{\partial \phi}{\partial x} + e \frac{\partial \phi}{\partial y} + f \phi - g \right) \equiv H$$

The same development applies to quasi-linear equations, in which a , b and c may also depend on ϕ and on its first derivatives.

Characteristics

$$a \frac{\partial^2 \phi}{\partial x^2} + b \frac{\partial^2 \phi}{\partial x \partial y} + c \frac{\partial^2 \phi}{\partial y^2} = - \left(d \frac{\partial \phi}{\partial x} + e \frac{\partial \phi}{\partial y} + f \phi - g \right) \equiv H$$

Let ϕ and its first derivatives be prescribed along a curve C .

Can the second derivatives fail to be determined by that information?

Writing the 3 available relations as a linear system for the 3 second derivatives, **the answer is yes where its determinant vanishes.**

$$a (dy)^2 - b dx dy + c (dx)^2 = 0$$

Dividing by $(dx)^2$ and writing h for the slope dy/dx gives a quadratic.

$$a h^2 - b h + c = 0, \quad h \equiv \frac{dy}{dx}$$

The discriminant

$$a h^2 - b h + c = 0, \quad h \equiv \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

The quantity under the root fixes how many **real characteristic directions pass through a point**, and therefore the type of the equation.

$$b^2 - 4ac \begin{cases} > 0 & \text{hyperbolic: two real characteristic families} \\ = 0 & \text{parabolic: one real family} \\ < 0 & \text{elliptic: no real characteristics} \end{cases}$$

If a , b and c are not constant the type may change from point to point. That is exactly what happens in transonic flow.

The type does not depend on the coordinates

Introduce new coordinates $\xi(x, y)$ and $\eta(x, y)$.

$$A \frac{\partial^2 \phi}{\partial \xi^2} + B \frac{\partial^2 \phi}{\partial \xi \partial \eta} + C \frac{\partial^2 \phi}{\partial \eta^2} = H \left(\xi, \eta, \phi, \frac{\partial \phi}{\partial \xi}, \frac{\partial \phi}{\partial \eta} \right)$$

The mapping is admissible provided its Jacobian does not vanish.

$$J = \frac{\partial(\xi, \eta)}{\partial(x, y)} = \frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial y} - \frac{\partial \xi}{\partial y} \frac{\partial \eta}{\partial x} \neq 0$$

Transforming the equation by the chain rule, the discriminant of the transformed equation is:

$$B^2 - 4AC = (b^2 - 4ac) \left(\frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial y} - \frac{\partial \xi}{\partial y} \frac{\partial \eta}{\partial x} \right)^2 = (b^2 - 4ac) J^2$$

The factor J^2 is positive, so the sign is preserved:

The characteristics are the right coordinates

Each new coefficient is a quadratic form in the first derivatives of the new coordinates:

$$A = a \left(\frac{\partial \xi}{\partial x} \right)^2 + b \frac{\partial \xi}{\partial x} \frac{\partial \xi}{\partial y} + c \left(\frac{\partial \xi}{\partial y} \right)^2$$

$$B = 2a \frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial x} + b \left(\frac{\partial \xi}{\partial x} \frac{\partial \eta}{\partial y} + \frac{\partial \xi}{\partial y} \frac{\partial \eta}{\partial x} \right) + 2c \frac{\partial \xi}{\partial y} \frac{\partial \eta}{\partial y}$$

$$C = a \left(\frac{\partial \eta}{\partial x} \right)^2 + b \frac{\partial \eta}{\partial x} \frac{\partial \eta}{\partial y} + c \left(\frac{\partial \eta}{\partial y} \right)^2$$

Everything here is known once ξ and η are chosen. The question is how to choose them.

$A = 0$ exactly when the curves $\xi = \text{constant}$ are characteristics.

The same argument on C gives the curves $\eta = \text{constant}$.

Canonical forms

Taking characteristics as coordinates removes A , or C , or both:

Hyperbolic: 2 real families, take both, so $A = C = 0$

$$\frac{\partial^2 \phi}{\partial \xi \partial \eta} = H \left(\xi, \eta, \phi, \frac{\partial \phi}{\partial \xi}, \frac{\partial \phi}{\partial \eta} \right)$$

Parabolic: 1 family only, take it as η , so $C = 0$ and then $B = 0$

$$\frac{\partial^2 \phi}{\partial \xi^2} = H \left(\xi, \eta, \phi, \frac{\partial \phi}{\partial \xi}, \frac{\partial \phi}{\partial \eta} \right)$$

Elliptic: no real family; take real and imaginary parts, $B = 0$, $A = C$

$$\frac{\partial^2 \phi}{\partial \xi^2} + \frac{\partial^2 \phi}{\partial \eta^2} = H \left(\xi, \eta, \phi, \frac{\partial \phi}{\partial \xi}, \frac{\partial \phi}{\partial \eta} \right)$$

In each case H is a different function of the 5 arguments.

Hyperbolic equations

Two real characteristic families exist. With constant coefficients their slopes are constant:

$$\frac{dy}{dx} = \lambda_1, \quad \frac{dy}{dx} = \lambda_2$$

Each is an ordinary differential equation for a family of straight lines:

$$\int dy = \lambda_1 \int dx \implies y = \lambda_1 x + C_1, \quad y = \lambda_2 x + C_2$$

$$\boxed{\xi \equiv C_1 = y - \lambda_1 x, \quad \eta \equiv C_2 = y - \lambda_2 x}$$

ξ is constant along the first family and η along the second, so they are the natural coordinates.

Moving along a characteristic changes x and y , but leaves ξ (or η) fixed.

A second canonical form

Rotating the characteristic coordinates gives the other standard hyperbolic form:

$$\bar{\xi} = \frac{\xi + \eta}{2}, \quad \bar{\eta} = \frac{\xi - \eta}{2}$$

$$\frac{\partial^2 \phi}{\partial \bar{\xi}^2} - \frac{\partial^2 \phi}{\partial \bar{\eta}^2} = \bar{H} \left(\bar{\xi}, \bar{\eta}, \phi, \frac{\partial \phi}{\partial \bar{\xi}}, \frac{\partial \phi}{\partial \bar{\eta}} \right)$$

The model equation is the second-order wave equation, with wave speed c .

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Its slopes are $\pm 1/c$, so its characteristics are:

The wave equation in characteristic form

$$\xi = x + ct, \quad \eta = x - ct$$

$$\frac{\partial}{\partial t} = c \left(\frac{\partial}{\partial \xi} - \frac{\partial}{\partial \eta} \right), \quad \frac{\partial}{\partial x} = \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta}$$

Applying each operator twice and substituting into the wave equation, the two second derivatives combine:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = -4c^2 \frac{\partial^2 u}{\partial \xi \partial \eta} = 0 \quad \implies \quad \frac{\partial^2 u}{\partial \xi \partial \eta} = 0$$

This is the canonical form of a previous slide.

Integrating twice

Integrating in ξ , the inner derivative can depend on η only.

Integrating again in η gives:

$$\frac{\partial}{\partial \xi} \left(\frac{\partial u}{\partial \eta} \right) = 0 \implies \frac{\partial u}{\partial \eta} = F_2'(\eta)$$

$$u = F_1(\xi) + F_2(\eta) = F_1(x + ct) + F_2(x - ct)$$

We have 2 arbitrary functions, one running right and one running left, both at speed c .

The initial data fix them:

$$u(x, 0) = F_1(x) + F_2(x) = f(x)$$

$$\frac{\partial u}{\partial t}(x, 0) = c F_1'(x) - c F_2'(x) = g(x)$$

D'Alembert's solution

Integrate the second condition from 0 to x , add and subtract the first, and the two functions separate:

$$F_1(x) = \frac{f(x)}{2} + \frac{1}{2c} \int_0^x g(\tau) d\tau$$

$$F_2(x) = \frac{f(x)}{2} - \frac{1}{2c} \int_0^x g(\tau) d\tau$$

Evaluating F_1 at $x + ct$ and F_2 at $x - ct$ and adding:

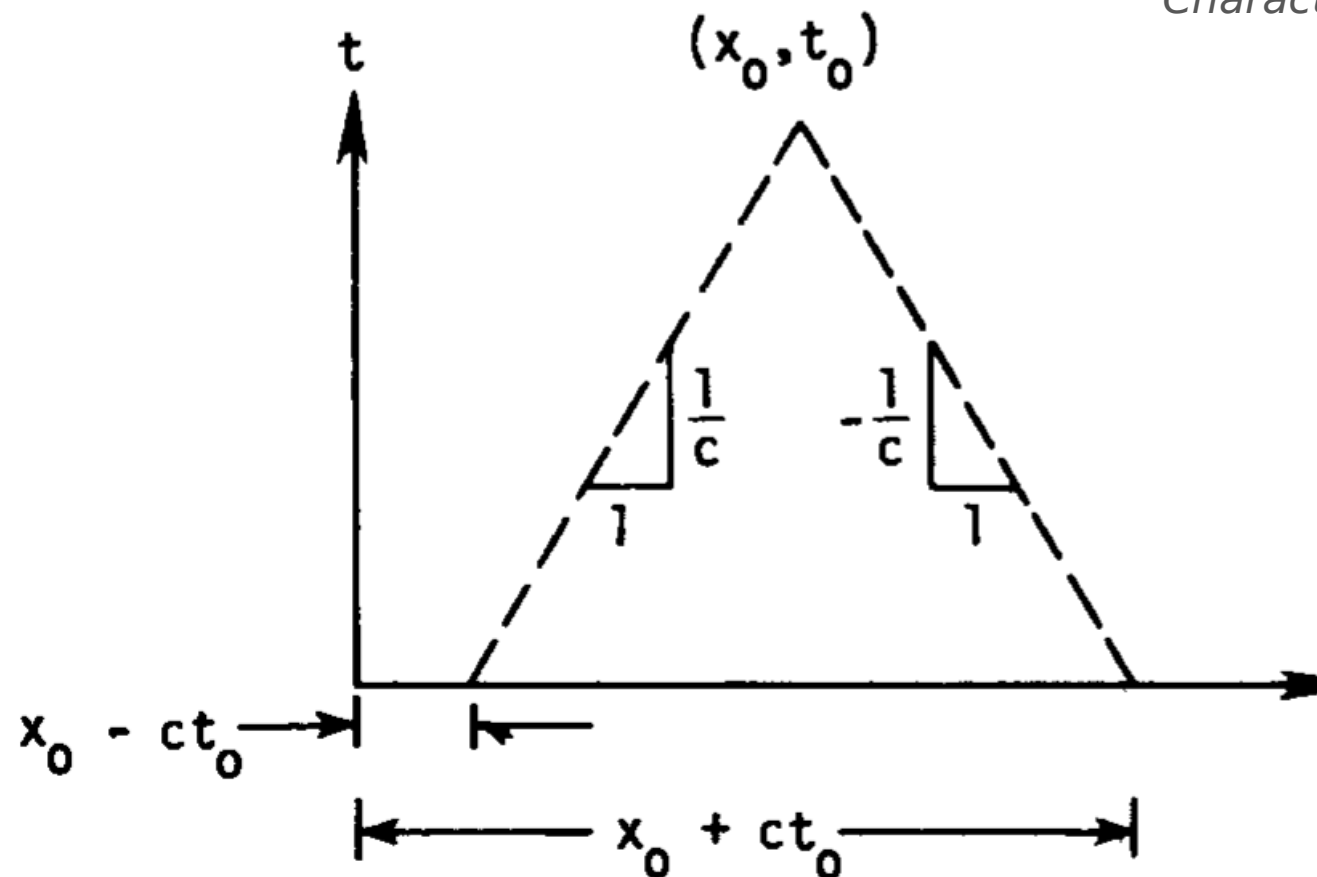
$$u(x, t) = \frac{f(x + ct) + f(x - ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tau) d\tau$$

The first term propagates the initial shape; the integral collects the initial velocity between the two characteristics.

Domain of dependence

The solution at a point depends only on the initial data lying between the two characteristics drawn back from it.

Characteristics for the wave equation.



$$x_0 - ct_0 \leq x \leq x_0 + ct_0$$

A disturbance outside that interval can never influence the point.

This is the property every explicit hyperbolic scheme has to respect.

Parabolic equations

The discriminant vanishes, the two characteristic slopes coincide, and only one family exists. Take it as η :

$$b^2 - 4ac = 0 \implies \frac{dy}{dx} = \frac{b}{2a} \implies \eta = y - \frac{b}{2a}x \implies C = 0$$

ξ is then any function independent of η , so that the Jacobian does not vanish.

B follows from the invariance identity:

$$B^2 - 4AC = (b^2 - 4ac) J^2 = 0, \quad C = 0 \implies B = 0$$

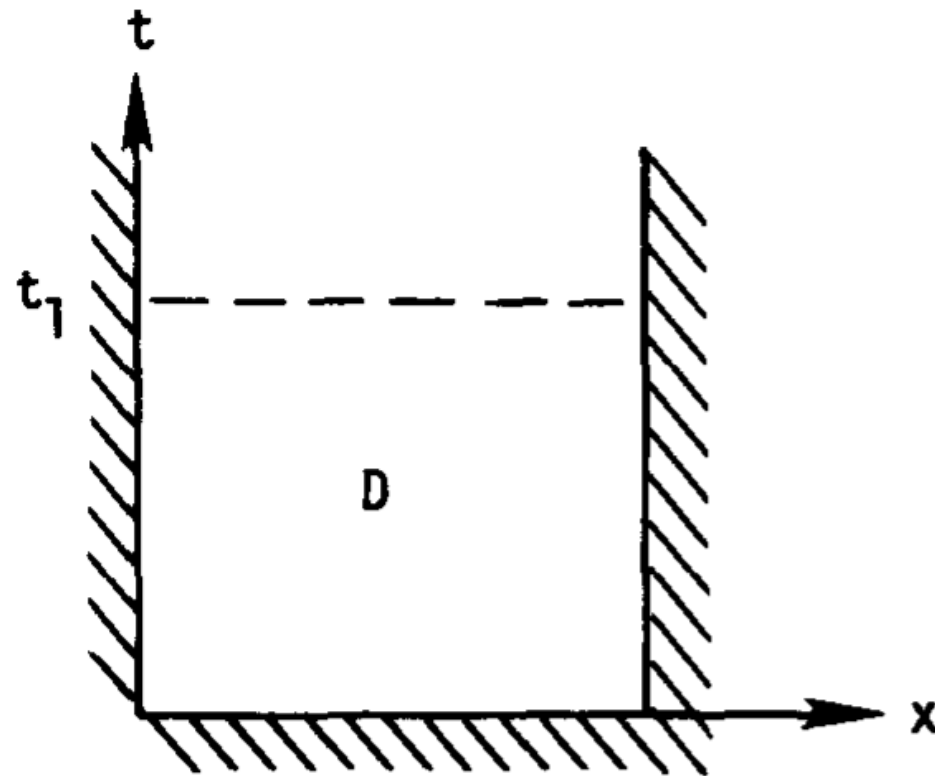
$$A \frac{\partial^2 \phi}{\partial \xi^2} = H \implies \frac{\partial^2 \phi}{\partial \xi^2} = \frac{H}{A}$$

One second derivative survives, and no second choice of coordinate was needed to remove the mixed term.

Parabolic equations: the heat equation

The model equation is the heat equation, with α the thermal diffusivity.

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$



Domain of dependence for a parabolic problem.

The solution at t_1 depends on the whole domain at every earlier time, and on both side boundaries.

Parabolic equations: diffusion

The Rayleigh problem: a flat plate, infinite, at rest in a viscous fluid. At $t = 0$ it is jerked into

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2} \quad \Longrightarrow \quad \delta \sim \sqrt{\nu t}$$

The 2 independent variables collapse into 1 similarity variable, and the PDE becomes an ODE:

$$\eta = \frac{y}{2\sqrt{\nu t}}, \quad f(\eta) = \frac{u}{U} \quad \Longrightarrow \quad \frac{d^2 f}{d\eta^2} + 2\eta \frac{df}{d\eta} = 0$$

$$\frac{df}{d\eta} = C_1 e^{-\eta^2} \quad \Longrightarrow \quad f(\eta) = 1 - \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-s^2} ds$$

$$u = U [1 - \operatorname{erf}(\eta)]$$

Hyperbolic: a signal arrives.

Parabolic: the effect is immediate everywhere but exponentially small far away.

Elliptic equations

The discriminant is negative: the characteristic slopes are a complex conjugate pair and no real characteristic exists.

$$\frac{dy}{dx} = \alpha \pm i\beta, \quad \alpha = \frac{b}{2a}, \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}$$

$$\xi = y - \alpha x, \quad \eta = \beta x$$

Take the real and imaginary parts of the complex characteristic coordinates.

$$A = C = a\beta^2 \neq 0, \quad B = 0 \quad \implies \quad \frac{\partial^2 \phi}{\partial \xi^2} + \frac{\partial^2 \phi}{\partial \eta^2} = \frac{H}{a\beta^2}$$

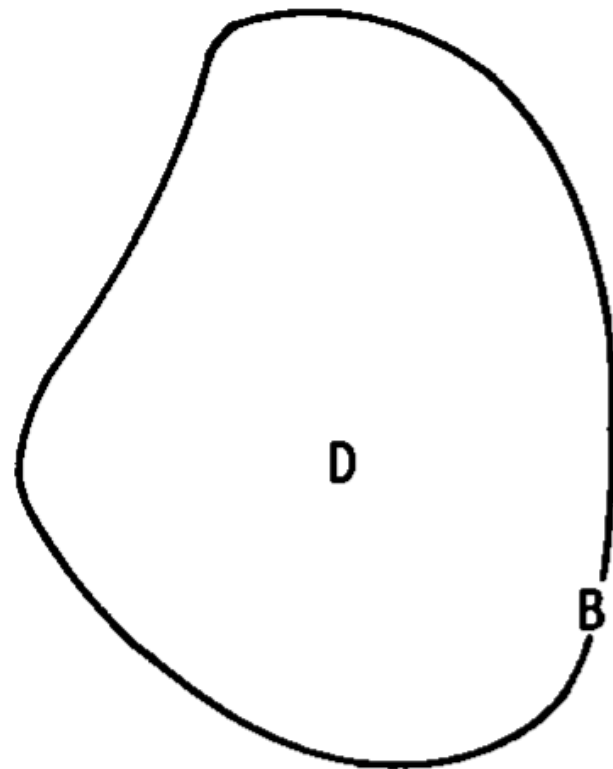
The canonical form is Laplace's operator.

Every elliptic problem looks like Laplace's equation.

Elliptic equations: Laplace's equation

The model equation is Laplace's equation.

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$



$$\begin{aligned} \nabla^2 u &= 0 && \text{IN } D \\ u &= f(s) && \text{ON } B \end{aligned}$$

The Dirichlet problem.

With no characteristics there is no preferred direction: **the solution at a point depends on the entire boundary.**

The well-posed problem

Hadamard:

A problem is well posed if the solution exists, is unique, and depends continuously on the initial and boundary data.

Uniqueness fails:

When initial data for a hyperbolic equation are prescribed along a characteristic.

Continuity fails:

When an elliptic equation is given data on part of the boundary only; small data then produce unbounded solutions.

An **ill-posed problem** cannot be repaired by a better numerical method.

Types of boundary data

Three specifications, named after the classical problems for Laplace's equation, with s being the arc length along the boundary.

$$\text{Dirichlet: } u = f(s)$$

$$\text{Neumann: } \frac{\partial u}{\partial n} = g(s)$$

$$\text{Robin: } a_1(s) \frac{\partial u}{\partial n} + a_2(s) u = h(s)$$

The names are used for any PDE.

Dirichlet data means the unknown itself is prescribed

Neumann data means its normal derivative is prescribed.

Systems of first-order equations

A physical process is rarely governed by a single equation, and a higher-order equation can usually be rewritten as a first-order system.

$$\frac{\partial \mathbf{u}}{\partial t} + [A] \frac{\partial \mathbf{u}}{\partial x} + [B] \frac{\partial \mathbf{u}}{\partial y} + \mathbf{r} = 0$$

Here $\mathbf{u}$ is the column of unknowns, and $[A]$ and $[B]$ are coefficient matrices.

$$\det ([A] - \lambda [I]) = 0$$

Hyperbolic in (x, t) : if the eigenvalues of $[A]$ are all real and distinct.

Elliptic in (x, t) : if the eigenvalues of $[A]$ are all complex.

The eigenvalues are the characteristic speeds

Let λ be a real eigenvalue of $[A]$ and ℓ its left eigenvector.

$$\ell^T [A] = \lambda \ell^T$$

$$\ell^T \left(\frac{\partial \mathbf{u}}{\partial t} + \lambda \frac{\partial \mathbf{u}}{\partial x} \right) = 0 \quad \text{along} \quad \frac{dx}{dt} = \lambda$$

The combination $\ell^T u$ then changes only along the line $dx/dt = \lambda$.

That line is a characteristic.

n real and distinct eigenvalues give **n** such directions, one for each unknown, and the system can be solved by marching.

The wave equation as a system

Define two new variables, and the second-order wave equation becomes a pair of first-order equations.

$$v = \frac{\partial u}{\partial t}, \quad w = c \frac{\partial u}{\partial x} \quad \Longrightarrow \quad \frac{\partial v}{\partial t} = c \frac{\partial w}{\partial x}, \quad \frac{\partial w}{\partial t} = c \frac{\partial v}{\partial x}$$

$$\frac{\partial}{\partial t} \begin{bmatrix} v \\ w \end{bmatrix} + \begin{bmatrix} 0 & -c \\ -c & 0 \end{bmatrix} \frac{\partial}{\partial x} \begin{bmatrix} v \\ w \end{bmatrix} = 0$$

The eigenvalues of the coefficient matrix are the characteristic slopes we already found for the wave equation.

$$\lambda^2 - c^2 = 0 \quad \Longrightarrow \quad \lambda_{1,2} = \pm c \quad \text{that is} \quad \left(\frac{dx}{dt} \right)_{1,2} = \pm c$$

The Cauchy-Riemann equations

Laplace's equation can also be replaced by a system of two first-order equations in two unknowns.

$$\frac{\partial u}{\partial x} = +\frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

These are the Cauchy-Riemann equations.

In matrix form, the eigenvalues of the coefficient matrix are imaginary.

$$\frac{\partial}{\partial x} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \frac{\partial}{\partial y} \begin{bmatrix} u \\ v \end{bmatrix} = 0 \quad \implies \quad \lambda_{1,2} = \pm i$$

Complex eigenvalues, so the system is elliptic, consistent with Laplace's equation.

Model equations: waves

First-order linear wave equation, or advection equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

Inviscid Burgers equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$$

Burgers' equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

Model equations: mixed type and equilibrium

Tricomi equation

$$y \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Poisson equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

Helmholtz equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + k^2 u = 0$$

The Tricomi equation is elliptic for $y > 0$ and hyperbolic for $y < 0$: a model of inviscid transonic flow.

Model equations: transport and dispersion

Advection-diffusion equation

$$\frac{\partial \xi}{\partial t} + u \frac{\partial \xi}{\partial x} = \alpha \frac{\partial^2 \xi}{\partial x^2}$$

Korteweg-de Vries equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0$$

Biharmonic equation

$$\frac{\partial^4 u}{\partial x^4} + 2 \frac{\partial^4 u}{\partial x^2 \partial y^2} + \frac{\partial^4 u}{\partial y^4} = 0$$

Telegraph equation

$$\frac{\partial^2 u}{\partial t^2} + a \frac{\partial u}{\partial t} + b u = c^2 \frac{\partial^2 u}{\partial x^2}$$

Classification and solution strategy

Marching problems:

They are solved by **stepping**: the new level is computed from the old one, and only a few levels are stored.

Equilibrium problems:

They are solved by **relaxation or by one large algebraic system**, with every boundary acting at once.

The discretisation must respect the directions in which information travels.

Example: The upwinding method is meaningful only where real characteristics exist.

We need to classify the equation (PDE) first.

Choose the discretisation second.

Key equations

The general second-order equation, and its characteristic directions

$$\frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$

The transformed equation: the type is carried by A , B and C alone

$$A \frac{\partial^2 \phi}{\partial \xi^2} + B \frac{\partial^2 \phi}{\partial \xi \partial \eta} + C \frac{\partial^2 \phi}{\partial \eta^2} = H \left(\xi, \eta, \phi, \frac{\partial \phi}{\partial \xi}, \frac{\partial \phi}{\partial \eta} \right)$$

A first-order system: the eigenvalues of the coefficient matrix

$$\det ([A] - \lambda[I]) = 0$$

Two real characteristics, one, or none determine the type:
hyperbolic, parabolic, elliptic, respectively.